



Deep Learning

Introduction

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https://github.com/safayani/deep_learning_course



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Why Do We Need Neural Networks?

- The Limits of Traditional Programming
- **The Vision Challenge:**
  - Writing hand-crafted rules to recognize 3D objects under varying viewpoints, novel lighting, and cluttered backgrounds is practically impossible.
- **The “Black Box” of Human Cognition:**
  - We cannot write down an explicit algorithm for perception because we don’t consciously understand how our own brains process sensory input.
- **Intractable Complexity:**
  - Even if heuristic rules could be formulated, the resulting code would be horrendously complex and fragile.





The Machine Learning Paradigm

- From Explicit Rules to Data-Driven Programs

Traditional Programming	Machine Learning Approach
input: Data + Hand-written Rules →**Output:** Answers	**Input:** Data + Desired Outputs →**Output:** Learned Program / Model

- **Data-Driven Synthesis:**
  - Instead of coding logic by hand, we collect labeled examples $(X \rightarrow Y)$and let the learning algorithm synthesize the mapping function.
- **Non-Traditional Programs:**
  - The learned "program" consists of millions (or billions) of numerical parameters rather than human-readable code lines.
- **Generalization & Adaptability:**
  - **Generalization:** Performs accurately on novel, unseen data.
  - **Retrainability:** Adapts to shifting patterns simply by re-training on new data.
- **Economic Advantage:**
  - Scaling compute is now vastly cheaper than paying engineers to hand-craft task-specific heuristic programs





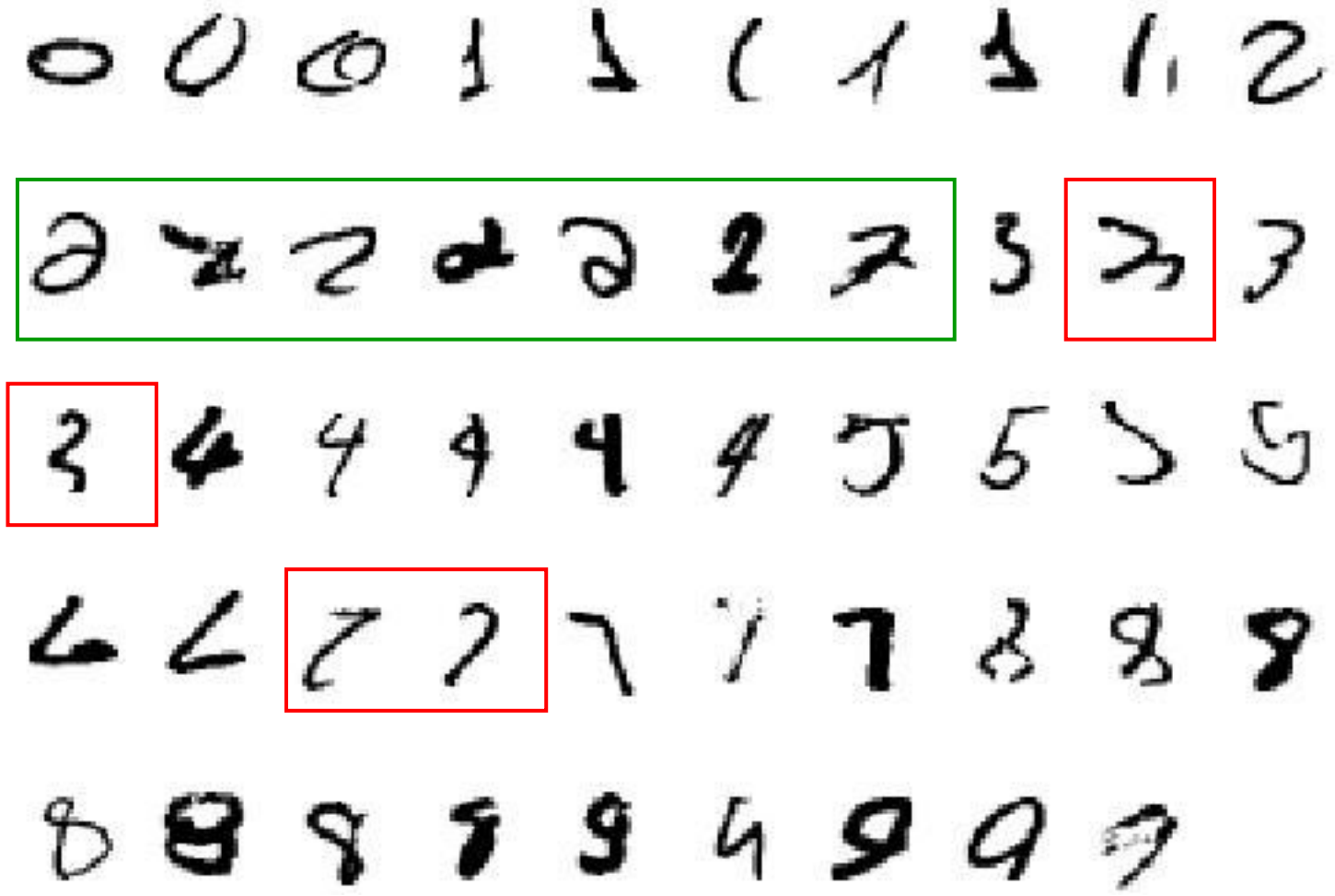
Standard Benchmark: The MNIST Dataset

- **What is MNIST?**
  - A canonical dataset of 28×28 pixel grayscale handwritten digits (0–9).
- **Why MNIST?**
  - **The Standard Model Organism:** Like fruit flies in genetics, it serves as the universal testing ground for learning algorithms.
  - **Accessible & Fast:** Publicly available and trainable in minutes on moderate hardware / neural networks.
  - **Extensively Benchmarked:** Provides a well-understood performance baseline across decades of machine learning methods.





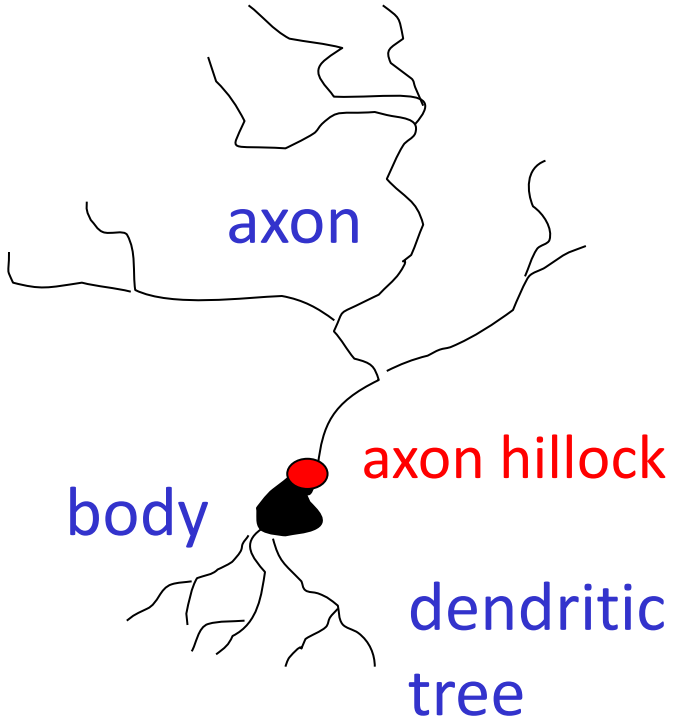
It is very hard to say what makes a 2



Neural Networks
What are neural networks?

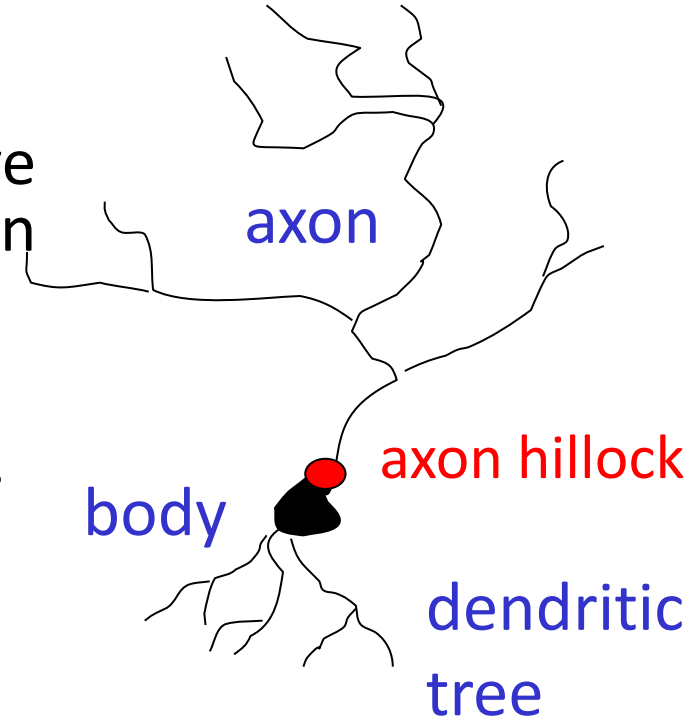
A Typical Cortical Neuron

- **1. Gross Physical Structure**
- **Dendritic Tree:** A branching structure that collects incoming inputs from thousands of other neurons.
- **Axon:** A single output transmission line that branches out to send signals to downstream cells.
- **Synaptic Contacts:** Axons connect to dendritic trees at junctions called **synapses**.
  - A spike arriving at a synapse injects electrical charge (depolarizing or hyperpolarizing) into the receiving cell.



A Typical Cortical Neuron

- **2. Integration & Spike Generation**
- **Charge Accumulation:** Incoming charges from all active synapses flow through the dendrites and accumulate in the cell body.
- **The Axon Hillock (The Comparator):**
  - Located at the junction between the cell body and the axon.
  - It continuously measures the accumulated charge against a critical **firing threshold** (typically around -55 mV).
- **All-or-None Firing:**
  - If accumulated charge $\geq$**Threshold (θ):** The neuron fires an outgoing spike (action potential) down the axon.
  - If accumulated charge $<$**Threshold (θ):** Nothing happens; the charge leaks away.



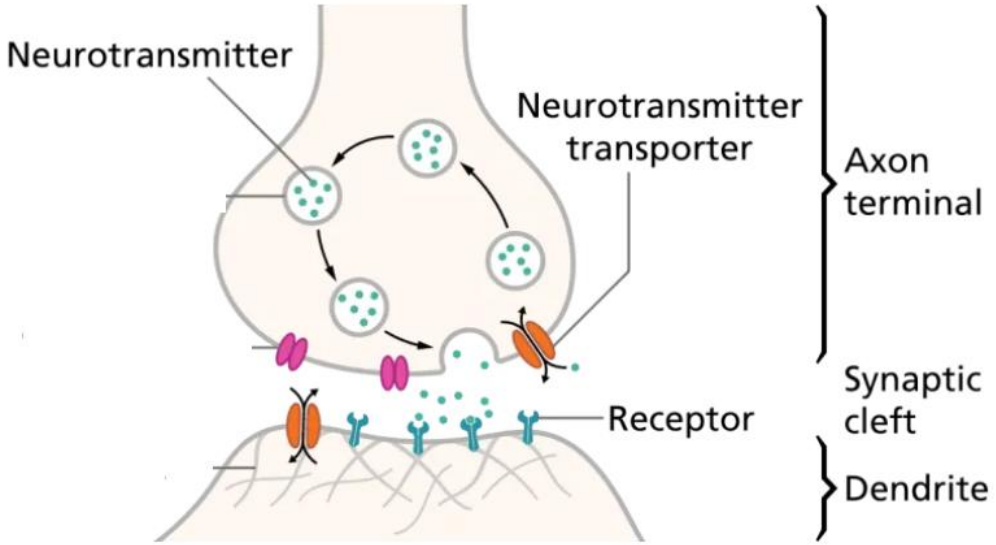
Synaptic Transmission

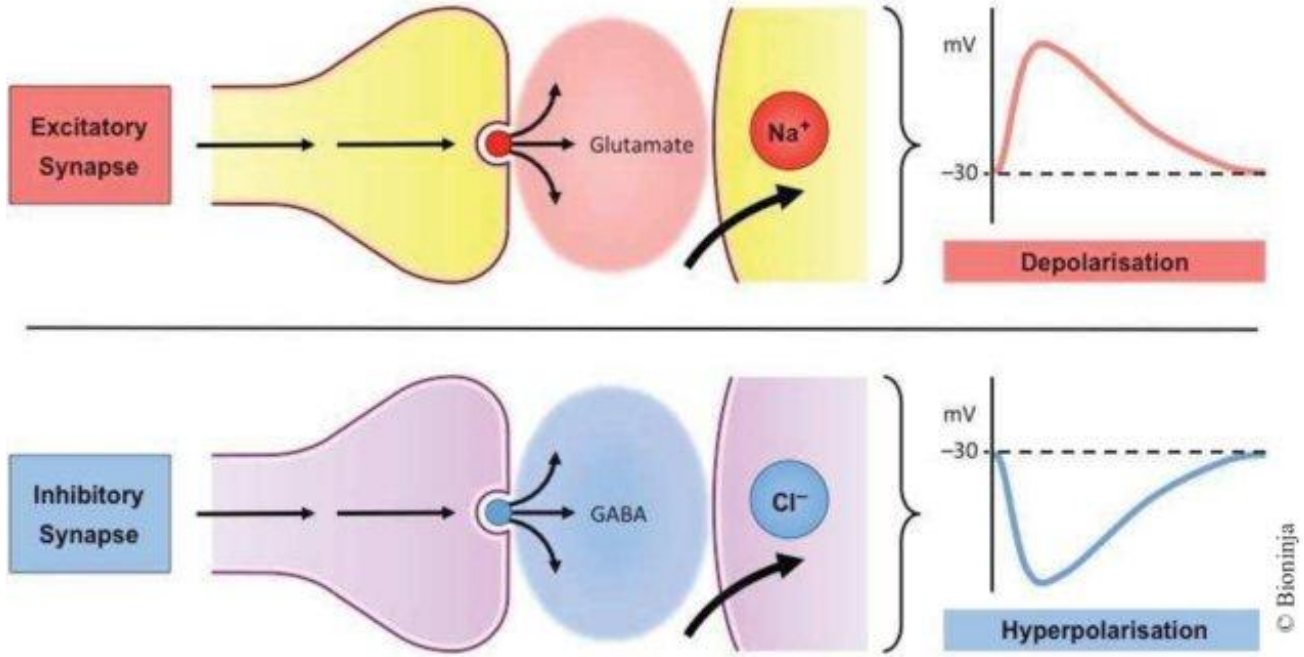
- **How Information Bridges the Gap Between Neurons**
- **Arrival of the Action Potential**
  - A voltage spike reaches the presynaptic axon terminal.
  - Triggers the release of chemical transmitters stored in synaptic vesicles.
- **Crossing the Synaptic Cleft**
  - Multiple classes of neurotransmitters diffuse across the microscopic junction ($\approx$ 20 nm).
- **Gating Ion Channels**
  - Transmitter molecules bind to target receptors on the postsynaptic membrane.
  - Receptors change shape, opening ion channels that alter the post-synaptic cell's electrical potential.
- **Signal Conversion:** Electrical Signal $\longrightarrow$ Chemical Diffusion $\longrightarrow$ Electrical Response





Synapses


Neurotransmitter
Neurotransmitter transporter
Axon terminal
Synaptic cleft
Receptor
Dendrite


Excitatory Synapse
Glutamate
Na+
mV
-30
Depolarisation
Inhibitory Synapse
GABA
Cl−
mV
-30
Hyperpolarisation
© Bioninja

How synapses adapt

- The effectiveness of the synapse can be changed:
  - vary the number of vesicles of transmitter.
  - vary the number of receptor molecules.

- Synapses are slow, but they have advantages over RAM
  - They are very small and very low-power.
  - They adapt using locally available signals
    - But what rules do they use to decide how to change?





How Synapses Adapt (Synaptic Plasticity)

- **1. Modifying Connection Strength (Synaptic Weight)**
- A synapse is not a fixed wire; its transmission strength can be turned up or down:
- **Sending Side (Presynaptic):**
  - Can change the **amount of transmitter** released per spike (more or fewer vesicles).
- **Receiving Side (Postsynaptic):**
  - Can increase or decrease the **number of receptor channels** embedded in its membrane.
- **Result:**
  - **Strong connection:** Releases more chemical →catches more ions →produces a large voltage jump.
  - **Weak connection:** Releases minimal chemical →fewer receptors →produces a negligible voltage jump.





How the Brain Works (On One Slide!)

1. Architecture & Signaling

- **Massive Interconnectivity:** Each neuron receives inputs from thousands of other neurons;
- **Digital Communication:** Cortical neurons communicate via discrete, all-or-nothing electrical pulses called **spikes** (action potentials).

2. Synaptic Weights as Computation

- **Synaptic Scaling:** The effect of each incoming spike is modulated by a **synaptic weight** (w_i).
- **Sign Matters:**
  - **Positive ($+w$):** Excitatory (depolarizes towards firing threshold).
  - **Negative ($-w$):** Inhibitory (hyperpolarizes away from threshold).
- **Plasticity & Learning:** Synaptic weights adapt over time so the network self-organizes to execute complex computations:
  - *Object recognition, speech comprehension, motor control, and planning.*

3. Biological Scale vs. Modern Hardware

- You have about 10^{11}**(**100 billion**)** neurons, each with an average of 10^{4}**(**$10{,}000$**)** synaptic connections.
- Total parameter count: $\sim 10^{14}$**to** 10^{15}**(hundreds of trillions)** of adaptive weights.
- **Massive Parallel Bandwidth:** Hundreds of trillions of weights operate in parallel within milliseconds, operating at roughly **20 Watts** of power—far outperforming modern workstations in energy efficiency and simultaneous throughput.

Modularity and the brain

- Different bits of the cortex do different things.
  - Local damage to the brain has specific effects.
  - Specific tasks increase the blood flow to specific regions.
- But cortex looks pretty much the same all over.
  - Early brain damage makes functions relocate.
- Cortex is made of general purpose stuff that has the ability to turn into special purpose hardware in response to experience.
  - This gives rapid parallel computation plus flexibility.
  - Conventional computers get flexibility by having stored sequential programs, but this requires very fast central processors to perform long sequential computations.

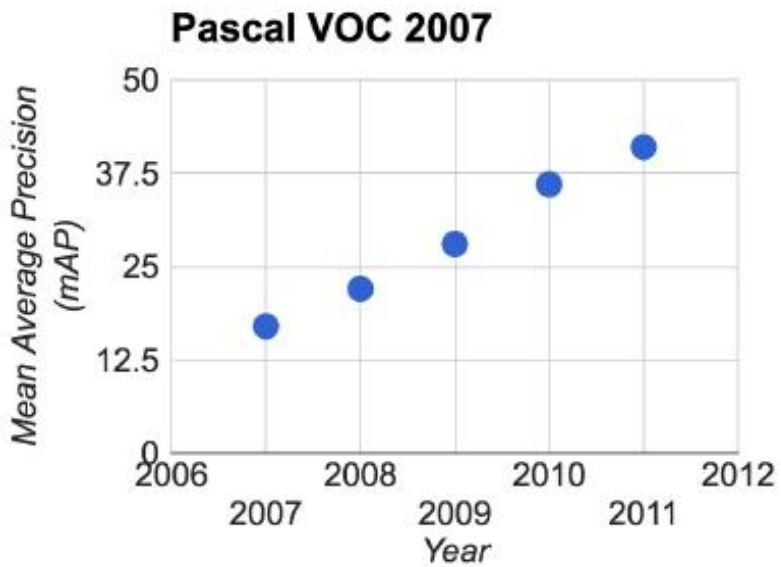




Object Recognition

PASCAL Visual Object Challenge
(20 object categories)
[Everingham et al. 2006-2012]







IMAGENET

www.image-net.org

22K categories and **14M** images

- Animals
  - Bird
  - Fish
  - Mammal
  - Invertebrate
- Plants
  - Tree
  - Flower
- Food
- Materials
- Structures
- Artifact
  - Tools
  - Appliances
  - Structures
- Person
- Scenes
  - Indoor
  - Geological Formations
- Sport Activities

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images



Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

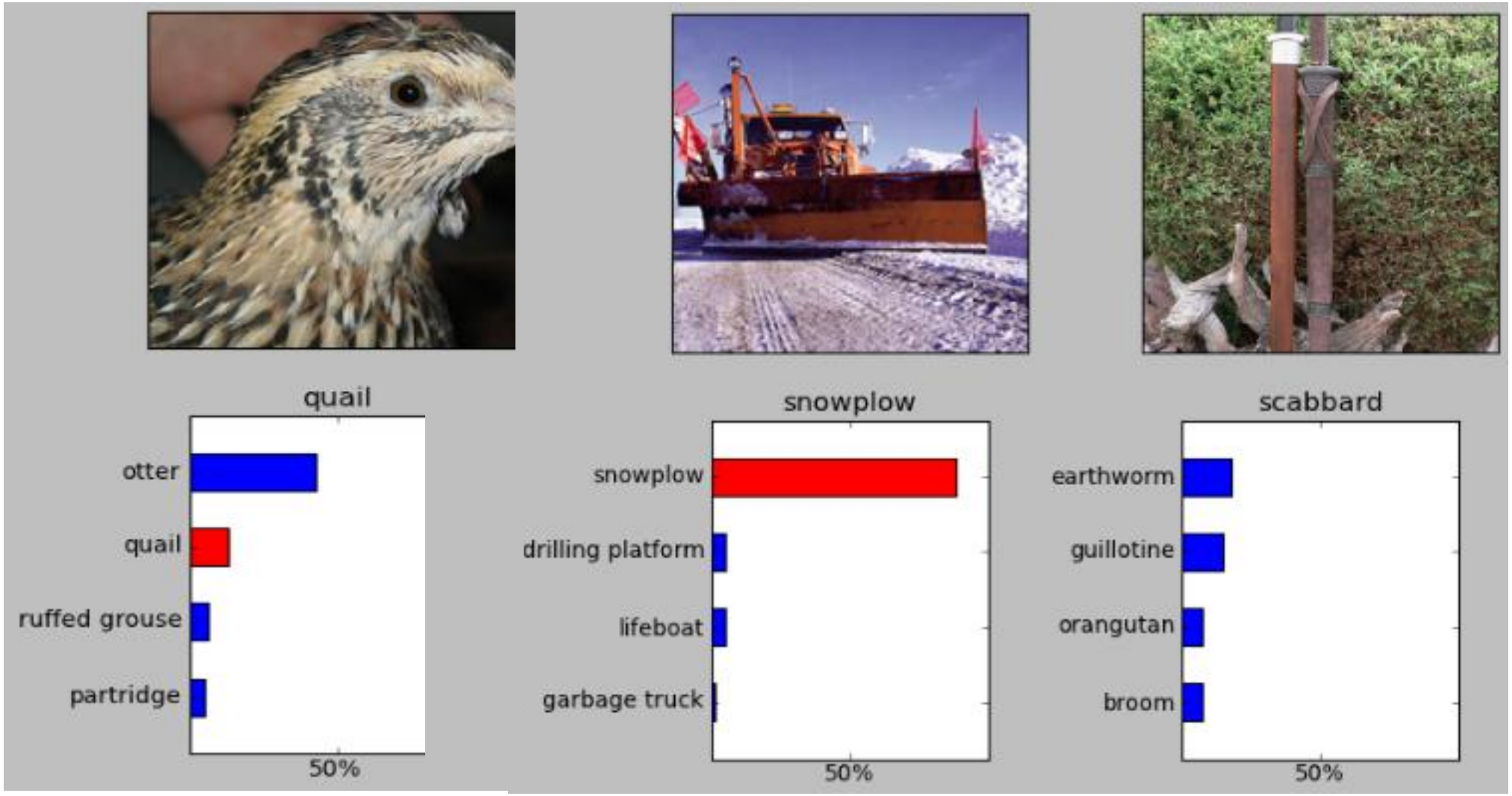
✓

Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle

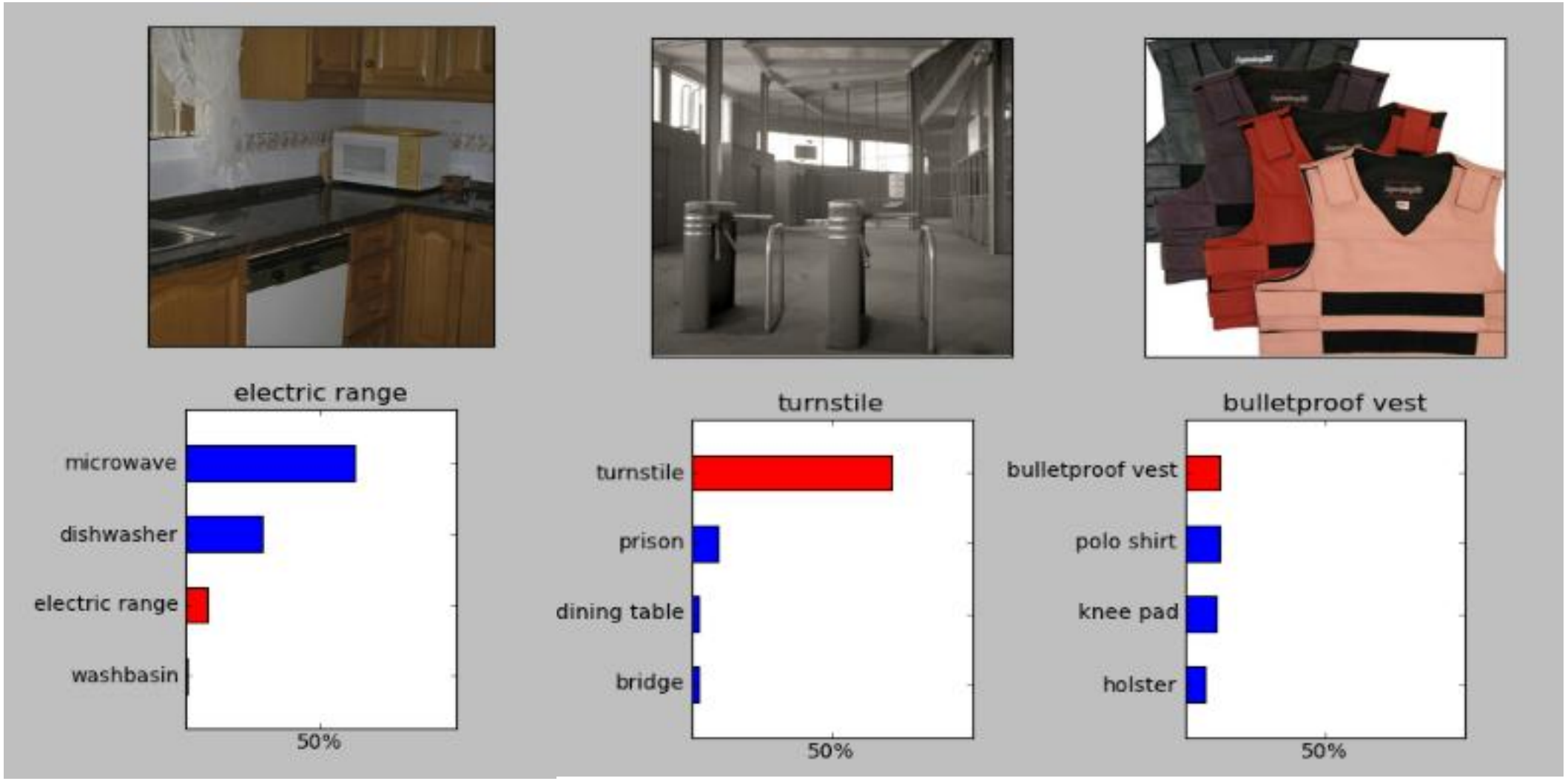
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Russakovsky et al. IJCV 2015

Some examples from an earlier version of the net

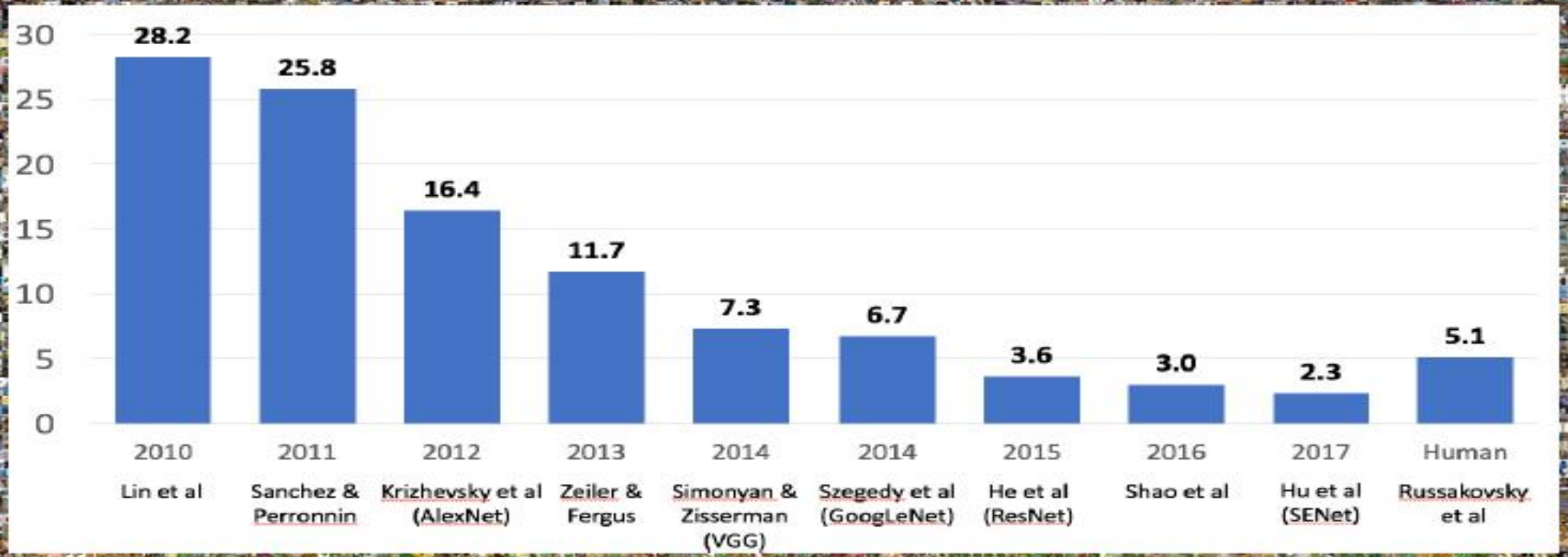


It can deal with a wide range of objects



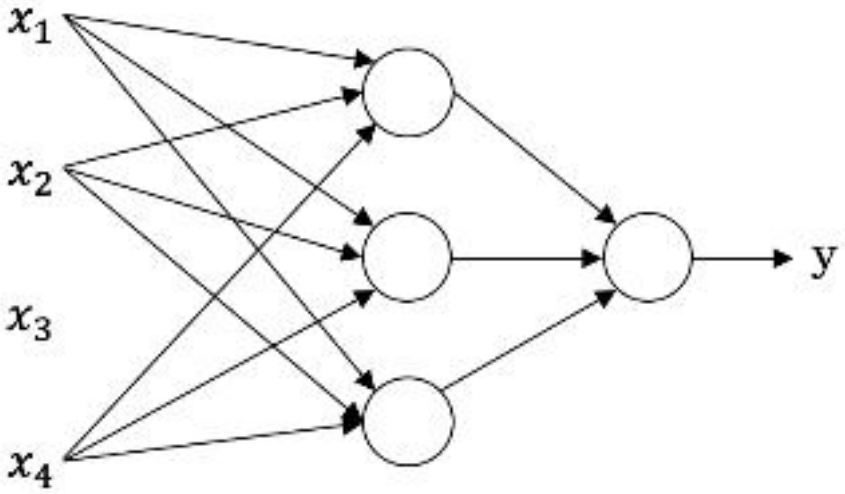
IMAGENET Large Scale Visual Recognition Challenge

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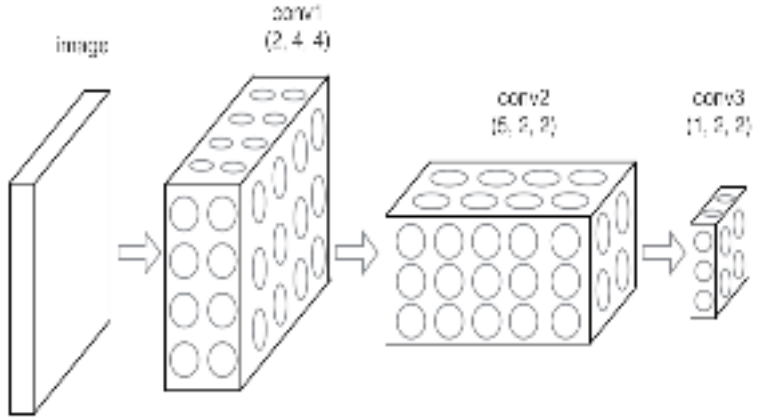


Russakovsky et al. IJCV 2015

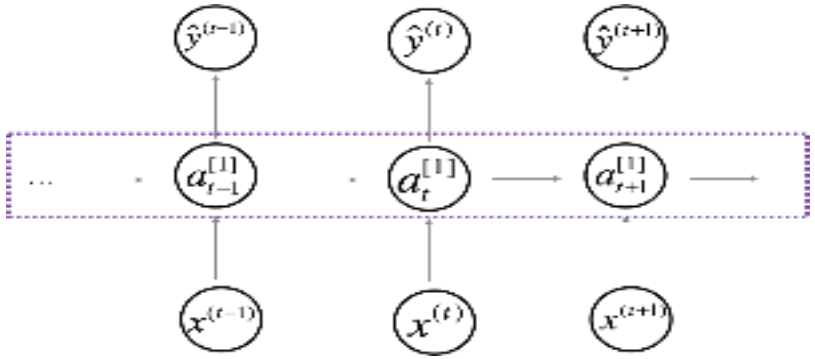
Neural Network examples



Standard NN



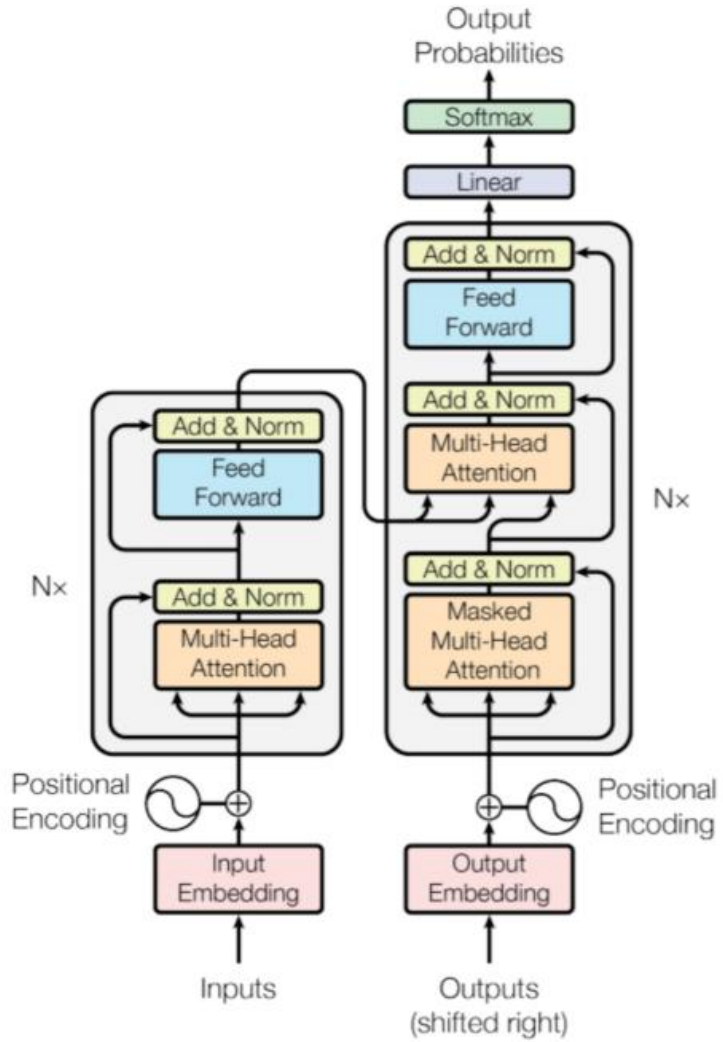
Convolutional NN



Recurrent NN

Neural Network examples

Transformers

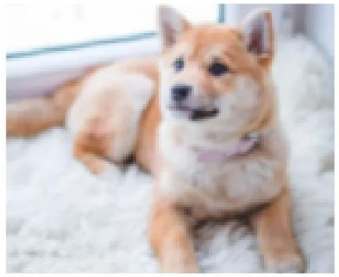


Some applications of Deep Learning

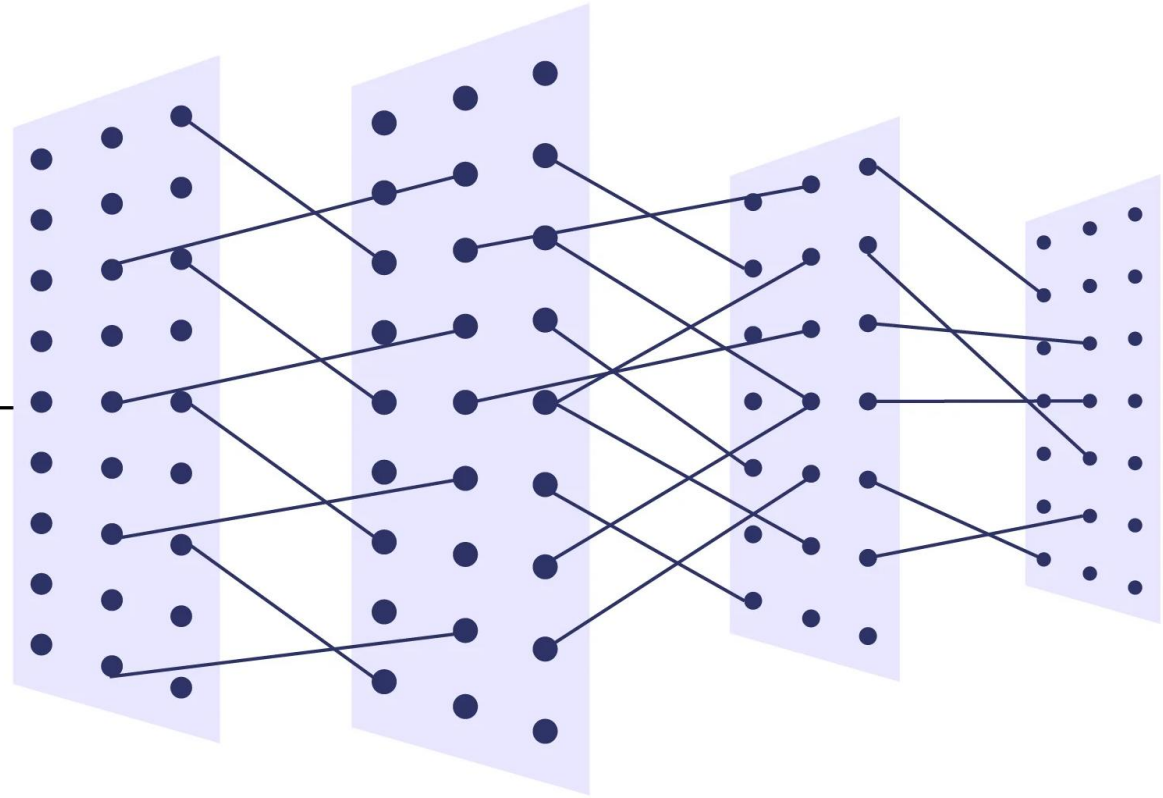
Object Classification



Cat

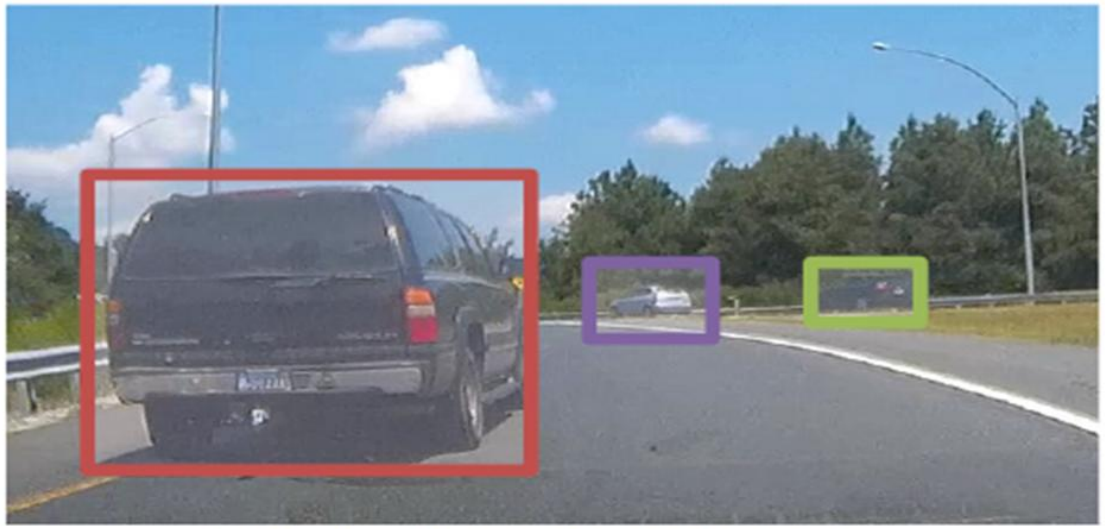


Dog



Output
Cat

Object Detection



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Person

Hammer

Neural style transfer




Content Style



Generated image




Content Style



Generated image

[Images generated by Justin Johnson]

Image Captioning: Example Results

Captions generated using neuraltalk2
All images are CC0 Public domain:
cat suitcase, cat tree, dog, bear,
surfers, tennis, giraffe, motorcycle



A cat sitting on a suitcase on the floor



A cat is sitting on a tree branch



A dog is running in the grass with a frisbee



A white teddy bear sitting in the grass



Two people walking on the beach with surfboards



A tennis player in action on the court



Two giraffes standing in a grassy field



A man riding a dirt bike on a dirt track

Visual Question Answering



Q: What endangered animal is featured on the truck?

A: **A bald eagle.**
A: A sparrow.
A: A humming bird.
A: A raven.



Q: Where will the driver go if turning right?

A: **Onto 24 ¾ Rd.**
A: Onto 25 ¾ Rd.
A: Onto 23 ¾ Rd.
A: Onto Main Street.



Q: Who is under the umbrella?

A: **Two women.**
A: A child.
A: An old man.
A: A husband and a wife.

Agrawal et al, "VQA: Visual Question Answering", ICCV 2015
Zhu et al, "Visual 7W: Grounded Question Answering in Images", CVPR 2016
Figure from Zhu et al, copyright IEEE 2016. Reproduced for educational purposes.

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Natural Language Processing

- Word Embedding
- Sentiment Classification
- Machine Translation
- Question answering, chat-gpt





Sentiment classification problem

x	y
The dessert is excellent.	★★★★☆
Service was quite slow.	★★☆☆☆
Good for a quick meal, but nothing special.	★★★☆☆
Completely lacking in good taste, good service, and good ambience.	★☆☆☆☆





Generative Models

- Generative Adversarial Network
- Variational Autoencoder





Image-to-Image Translation

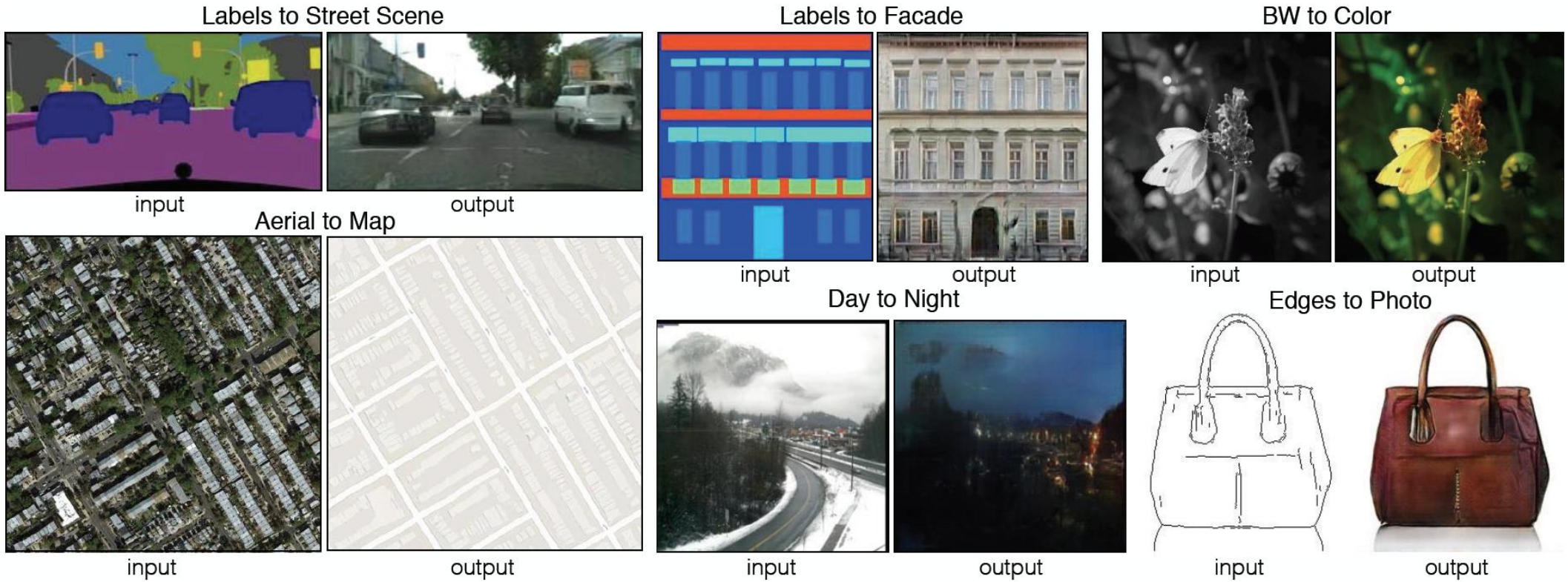


Figure 1 in the original paper.

[Link to an interactive demo of this paper]

Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. "**Image-to-image translation with conditional adversarial networks**". arXiv preprint arXiv:1611.07004. (2016).

Text-to-Image Synthesis

Motivation

Given a text description, generate images closely associated.

Uses a conditional GAN with the generator and discriminator being condition on “dense” text embedding.

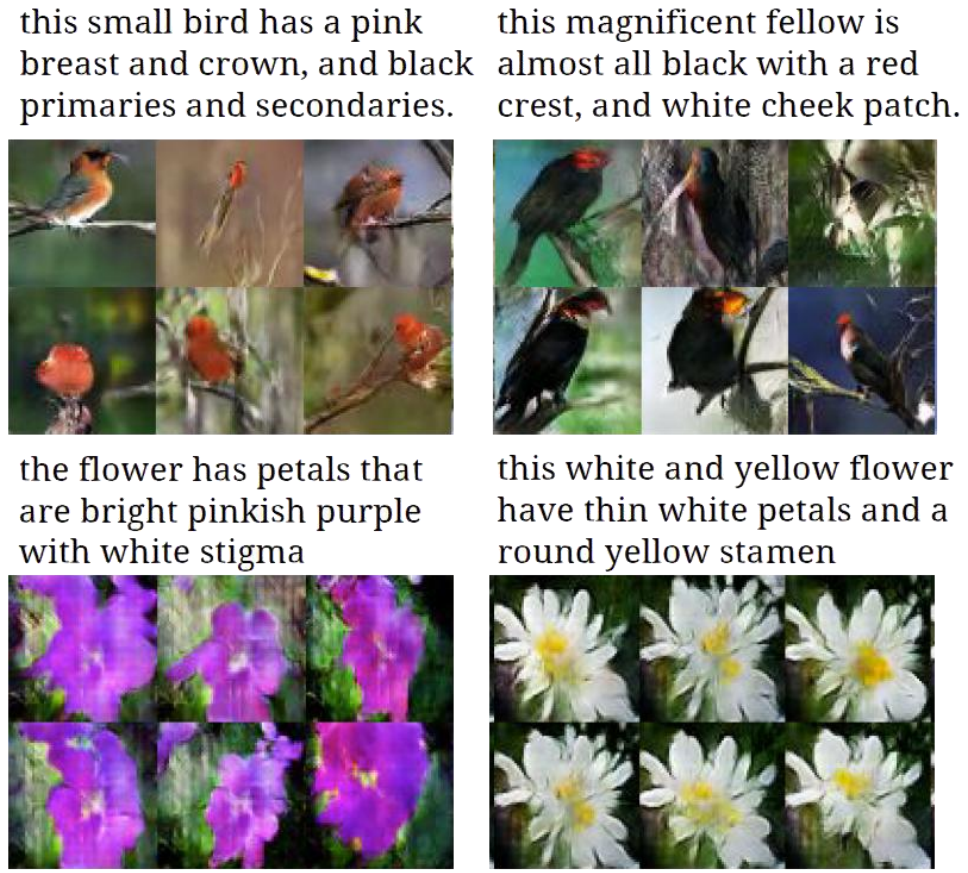
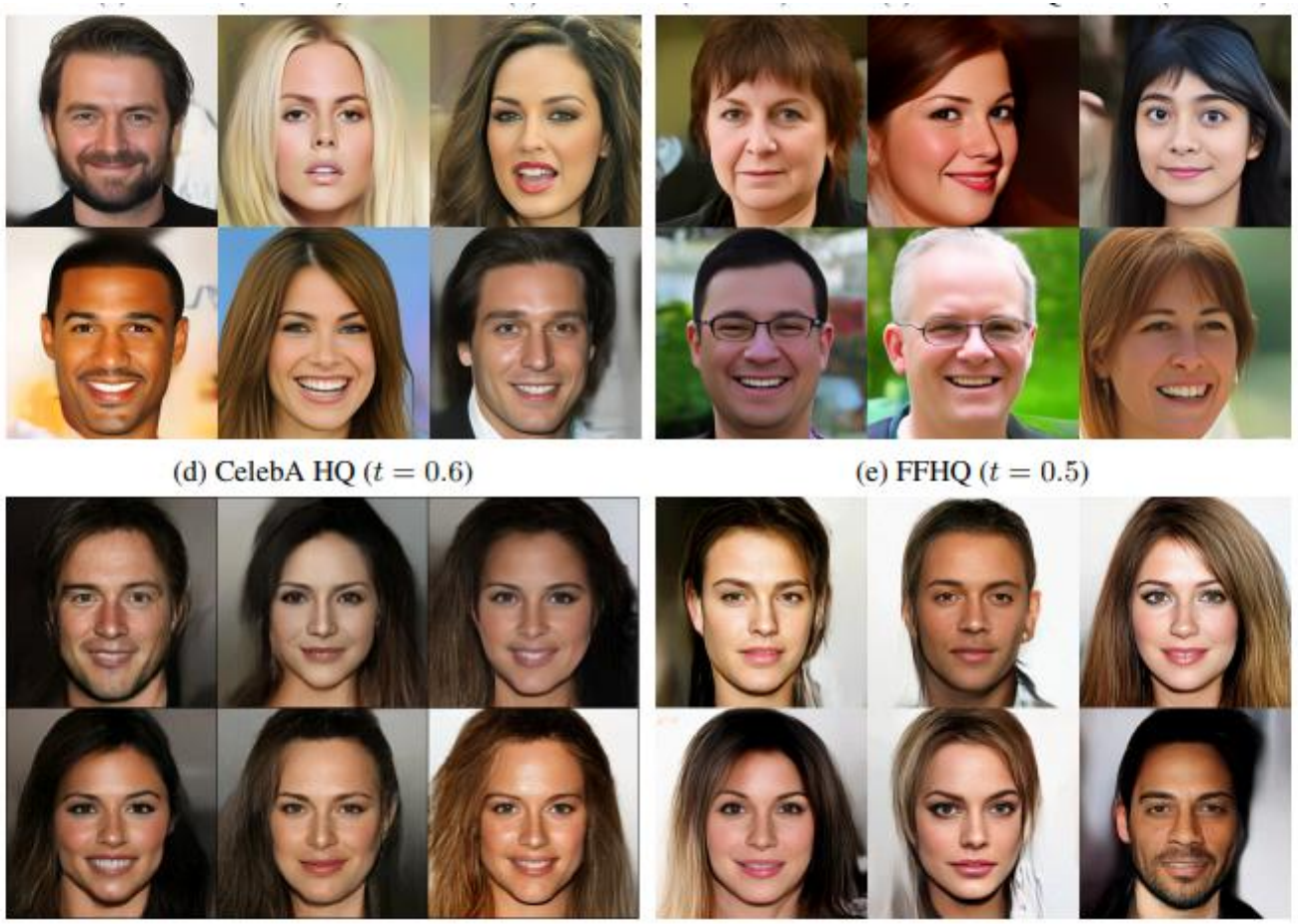


Figure 1 in the original paper.

Reed, S., Akata, Z., Yan, X., Logeswaran, L., Schiele, B., & Lee, H. “**Generative adversarial text to image synthesis**”. ICML (2016).

Face images generated with a Variational Autoencoder



(d) CelebA HQ ($t = 0.6$)

(e) FFHQ ($t = 0.5$)

A. Vahdat and J. Kautz, “NVAE: A deep hierarchical variational autoencoder,” in Proc. Conf. Neural Inf. Process. Syst., 2020, pp. 19667–19679

Course outline

- Introduction
- Logistic regression
- Multi-layer perceptron (MLP)
- Back Propagation (BP)
- Regularization concept
- Optimization
- Convolution NNs
- Introducing some famous Convolution NN architectures (LeNet-5, AlexNet, VGG, google net, Resnet, mobile net, efficient net)
- Some applications of deep learning(localization, detection, face recognition, style transfer)
- Recurrent neural network (GRU, LSTM)
- Transformers
- Basic concepts of NLP
- Generative AI (GAN, VAE)





Related Courses

- Deep Learning, Andrew NG.



- Convolutional Neural Networks for Visual Recognition, Fei-Fei Lee, Justin Johnson, Serena Yeung





- Neural Network for Machine Learning

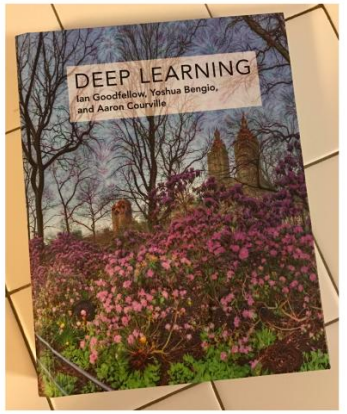








Textbooks



- Deep learning, I. Goodfellow, j. Bengio

MIT Press, 2016





- Neural Networks and deep learning, M. Nielsen

Determination Press, 2015



Course Strategy

- Assignments and project 5-7 marks (python)
- Midterm and quizzes 5-7 marks
- Final 5-7 marks







هوش مصنوعی برای همه

تنظیمات | چیدمان صفحه

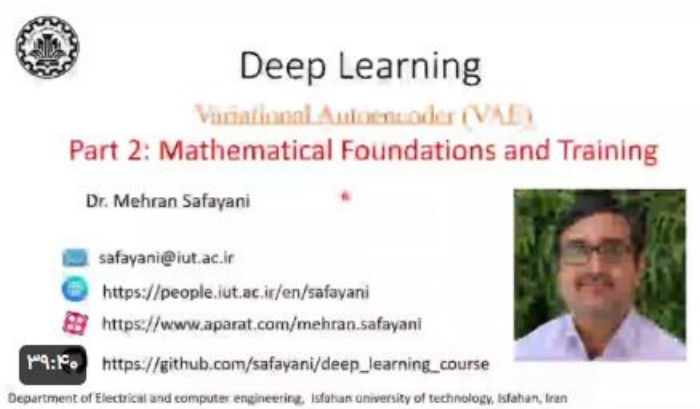
۸۹۳ دنبال‌کننده | ۸۱.۲ هزار بازدید کل

خانه | ویدیوها | ویدیوهای کوتاه | لیست‌های پخش | درباره کانال

یادگیر عمیق ج ۲۷ خودکدگذار تغییراتی (VAE) بخش ۲: مبانی ریاضی و آموزش ۱۴۰۵

چگونه مدل VAE را آموزش دهیم روابط آموزش VAE محاسبه ELBO Evidence Lower Bound تابع هدف حقه بازپارامترسازی Reparameterization trick مثال هایی از تولید داده

۱۲۷ بازدید ۴ ماه پیش



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